



Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  void factor(int num) {
4      int count = 0;
5      count++; // initialization of i = 1
6      for(int i = 1 ; i <= num ; i++) {
7          count++; // for the times of valid condition checks
8          count++; // for i++
9          if(num % i == 0) {
10             count ++; // for the count of times of execution of if condition
11             printf("%d ", i);
12         }
13     }
14     printf("%d", count);
15 }
16
17 int main() {
18     int n;
19     scanf("%d", &n);
20     factor(n);
21     return 0;
22 }
```

Check

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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